

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVC503 - STRUCTURAL MECHANICS - II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

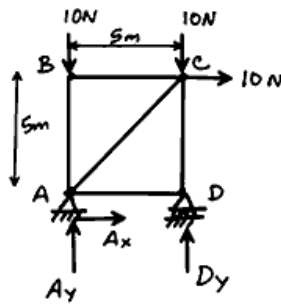
Q1. Objective type questions. (Compulsory Question)

12

- 1 A pin-jointed truss has 12 members, 7 joints, and 3 support reactions. The truss is:
A) Perfect B) Internally indeterminate
C) Externally unstable D) Kinetically unstable
Which of the following statements is true regarding an indeterminate truss?
A) It can be analyzed using only equilibrium equations.
B) Compatibility conditions are not required.
C) It requires additional equations derived from compatibility and material behavior.
D) It cannot carry any external load.
For which type of loading is the influence line concept most useful?
A) Uniformly distributed static load B) Dead load
C) Moving live loads D) Earthquake loads
- 4 The Müller-Breslau Principle is used to draw:
A) SFD B) BMD C) Influence line diagrams (ILD) D) Stress-strain curves
- 5 In a suspension bridge, the shape of the cable under its own weight is a:
a) Parabola b) Catenary c) Circle d) Ellipse
- 6 The flexibility matrix method is used for the analysis of...
a) Statically determinate structures b) Statically indeterminate structures
c) Both statically determinate and indeterminate structures d) None of the above
In a linear elastic structural element, stiffness is ----
a) Directly proportional to flexibility b) Equal to flexibility
c) Inversely proportional to flexibility d) Stiffness & flexibility are not related
The unit for stiffness is -----
a) N-m b) kN-mm c) N/m d) Both a & b
In finite element modeling every element connects to
a) 4 nodes b) 3 Nodes c) 2 Nodes d) Infinite No of Nodes
- 10 For truss analysis, which type of elements are used?
a) Triangle b) Bars c) Rectangle d) Parallelogram
- 11 The stiffness coefficient k_{ij} indicates -----
A) force at i due to a unit deformation at j
B) deformation at j due to a unit deformation at i
C) deformation at i due to a unit deformation at j
D) force at j due to a unit deformation at i
- 12 Basic unknowns in the flexibility matrix method are -----
a) Redundant forces b) Displacements of joints c) Both (a) & (b) d) None

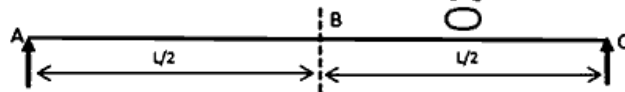
Q2. Solve the following.

- A) Calculate the external reactions and internal forces developed in the truss given below. Use the strain energy method. 6



Draw an influence line diagram (ILD) of the beam shown in the fig. below for 3 cases:

- i) ILD for reaction at Support A. ii) ILD for shear force at section B.
iii) ILD for bending moment at section B.



Q3. Solve the following.

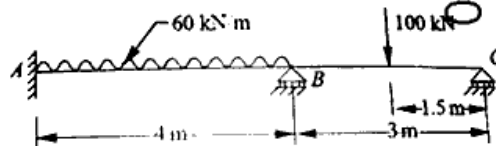
- A) Explain in detail the role of cables in a suspension bridge. 6

- B) Explain the two-hinged arch, three-hinged arch, and fixed arch with reference to their structural action. 6

Q4. Solve Any Two of the following.

Write the stepwise procedure to solve indeterminate structure by using the direct flexibility method of structural analysis. 6

Analyze the two-span continuous beam as shown in the figure and the final BMD by the flexibility matrix method. 6



- C) Explain the concept of Muller-Breslau's principle with a suitable example. 6

Q5. Solve Any Two of the following.

Define stiffness & write the procedure/steps to be adopted to solve any problem by using the stiffness/displacement method of analysis. 6

Develop the **stiffness matrix** for the cantilever beam of span L subjected to point load 'P' kN and clockwise moment 'M' at free end. 6

Determine the stiffness coefficients and develop the stiffness matrix for the cantilever beam of span L subjected to point load 'P' kN. 6

Q6. Solve Any Two of the following.

- A) Explain in detail the concept of 'shape function'. 6

- B) Explain the different type of finite elements. 6

- C) Explain in detail the concept of 'Pascal's Triangle.' 6

*** End ***